Decision Modeling Project: Elaboration of Decision Models for the Nutri-Score Label

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Abstract

This report details the elaboration of several decision models designed to evaluate food products by combining nutritional quality and environmental impact. Building upon the existing Nutri-Score system and the Green-Score, we first constructed a balanced database of over 200 products using an iterative web scraping approach. We then implemented the standard Nutri-Score algorithm with a graphical user interface. To explore new classification methodologies, we developed three distinct models: a Multi-Criteria Decision Analysis (MCDA) model using ELECTRE TRI, a compensatory Weighted Sum Model (WSM), and a Machine Learning classifier based on Random Forest. We defined limiting profiles using statistical quantiles and compared the results of each model against the traditional Nutri-Score. Our analysis concludes that while Machine Learning offers superior accuracy in replicating existing patterns, the ELECTRE TRI method provides the transparency and non-compensatory safety required for public health labeling.

Contents

1 Introduction and Context

1.1 What is the Nutri-Score?

The Nutri-Score is a nutrition labeling system that converts the nutritional value of food products into a simple code consisting of 5 letters, each associated with its own color. Elaborated by public health authorities, this logo allows consumers to distinguish at a glance which products are recommended and which should be avoided.

The score is awarded based on a scientific algorithm that takes into account, per 100g or 100ml of product:

- **Nutrients to limit (Negative Component N):** energy value, sugars, saturated fats, and salt.
- **Nutrients to encourage (Positive Component P):** fiber, protein, fruits, vegetables, and nuts.

The general calculation formula involves subtracting the positive points from the negative points. The lower the final score, the more favorable the product's nutritional profile. A threshold table then classifies the product from **A** (Dark Green) to **E** (Red).

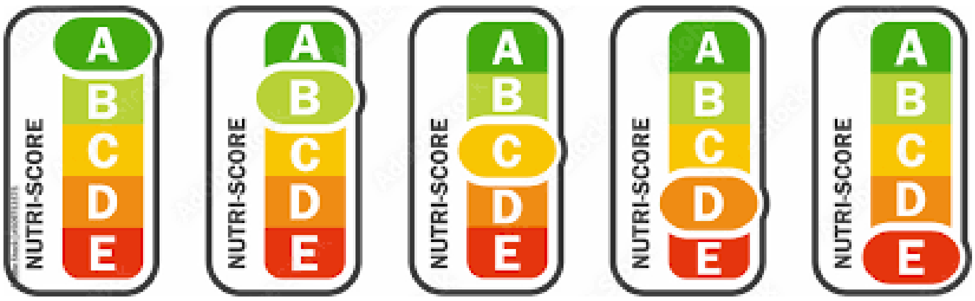

Figure 1: The Nutri-Score logo and its 5 classes

In parallel, the project integrates the **Green-Score**, an indicator that classifies products from A to E according to their environmental impact (Life Cycle Assessment, transport, packaging).

2 Database Construction

2.1 Data Source

In accordance with the project requirements, we elaborated our own set of alternatives (food products). We chose to utilize public data from the **OpenFoodFacts** platform, which lists nutritional information for thousands of products worldwide.

2.2 Methodology: Iterative Scraping and Balancing

To construct a relevant database that strictly adheres to the imposed statistical constraints, we did not simply download a static dataset. Instead, we implemented an **iterative Web Scraping** procedure in Python.

The objective was to satisfy three major constraints simultaneously:

1. Obtain a minimum of **200 products**.
2. Ensure that each Nutri-Score label (A, B, C, D, E) represents at least **15%** of the total.
3. Ensure that each Green-Score label (A, B, C, D, E) represents at least **10%** of the total.

2.2.1 Collection Algorithm

Our approach was conducted in several iterative steps to correct natural market imbalances (where extreme classes like 'A' or 'E' might be under-represented depending on the food category):

- **Step 1: Initial Scraping.** We launched a first data extraction on a broad category (e.g., biscuits or ready-made meals) to obtain an initial batch of products.
- **Step 2: Distribution Analysis.** After each extraction, a script analyzed the class distribution. We frequently observed that certain classes were under-represented compared to others.
- **Step 3: Targeted Scraping (Readjustment).** To compensate for these gaps, we re-ran the scraper with specific parameters, targeting products likely to belong to the missing classes. For example, if class 'E' was missing, we targeted products rich in saturated fats; if class 'A' was missing, we targeted fiber-rich products.
- **Step 4: Validation Loop.** This process was repeated iteratively. In every loop, we added more data specifically for the factors that were deficient in the previous iteration. The loop continued until the addition of new data allowed us to satisfy the percentage thresholds for all classes, with a final total exceeding 200 products.

This method allowed us to build a robust and balanced dataset, avoiding biases that could have skewed the learning and evaluation of our decision models.

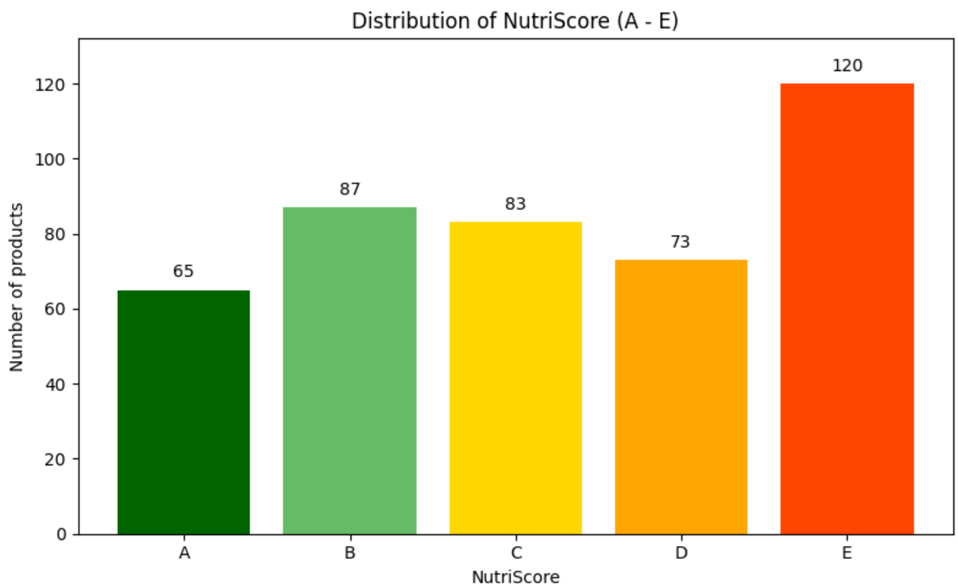

Figure 2: Distribution of NutriScore A–E

3 Implementation of the Standard Nutri-Score Algorithm

3.1 Algorithmic Implementation and GUI

Before developing our advanced decision models, we first implemented the standard Nutri-Score calculation method as a baseline. As required by the project specifications, we developed a Graphical User Interface (GUI) to allow for individual product assessment.

The application was built using the Python `tkinter` library. It strictly follows the computation rules provided by Santé Publique France and the project documentation:

- **Negative Component (N):** Inputs for Energy, Sugars, Saturated Fatty Acids, and Salt are converted into a score from 0 to 40 .
- **Positive Component (P):** Inputs for Fibers, Proteins, and Fruits/Vegetables are converted into a score from 0 to 15.

3.2 Calculation Logic and Exception Handling

The internal engine (`NutriScoreEngine` class) handles the specific conditional logic of the Nutri-Score. Specifically, it implements the rule regarding protein inclusion: if the Negative score is high (≥ 11) and the fruit content is low ($\leq 80\%$), the points for proteins are excluded from the calculation .

$$\text{Final Score} = \text{Total N} - \text{Total P} \tag{1}$$

The tool provides immediate visual feedback, changing the background color and displaying the Nutri-Score letter (A to E) corresponding to the calculated score (Figure 3).

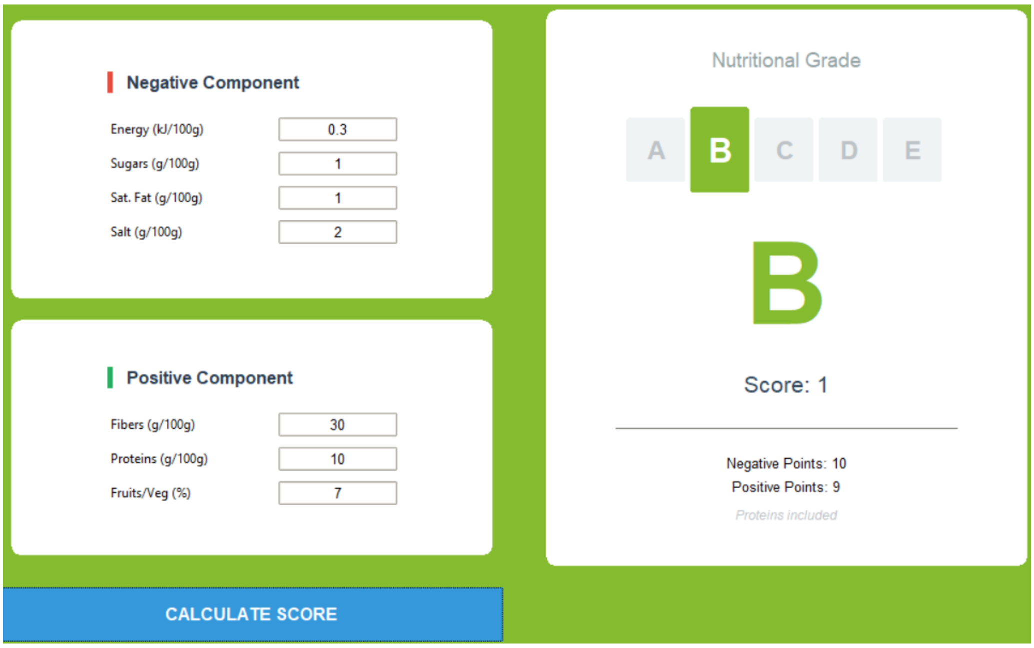

Figure 3: The Graphical User Interface (GUI) developed to calculate the Nutri-Score for individual products.

4 ELECTRE TRI Model: Implementation and Results

4.1 Methodology and Configuration

To move beyond the limitations of the standard Nutri-Score algorithm, we modeled the decision problem using ELECTRE TRI. This is a Multi-Criteria Decision Analysis (MCDA) method designed specifically for sorting alternatives into predefined ordered categories.

4.1.1 Criteria and Weights

We selected 8 decision criteria. Unlike the standard Nutri-Score, which uses a points-based system where bad scores can be compensated by good ones, ELECTRE treats each criterion individually. Crucially, we included the Green-Score as a full criterion to integrate the environmental dimension directly into the decision process.

- Negative Criteria (to minimize): Energy (kJ), Sugars (g), Saturated Fats (g), Salt (g).
- Positive Criteria (to maximize): Proteins (g), Fibers (g), Fruits/Vegetables (%), Green-Score.

Regarding the weighting strategy, we opted for an equal weighting approach ($w_i = 1/8$ for all criteria). In the absence of a specific decision-maker favoring nutrition over environment, this allows us to analyze the structural impact of the method itself without introducing subjective bias.

4.1.2 Determination of Limiting Profiles

A major challenge in applying ELECTRE TRI to a new dataset is defining the limiting profiles (π_k) that serve as boundaries between the categories (A', B', C', D', E'). Since there are no official profiles combining Nutrition and Ecology, we implemented a statistical quantile-based approach using our Python script. This ensures that our category boundaries reflect the actual market distribution of the products we collected.

The profiles were constructed as follows:

- π_2 (Boundary E'/D'): 20^{th} percentile
- π_3 (Boundary D'/C'): 40^{th} percentile
- π_4 (Boundary C'/B'): 60^{th} percentile
- π_5 (Boundary B'/A'): 80^{th} percentile

The direction of the profile values was automatically adjusted: increasing for positive criteria and decreasing for negative criteria.

4.2 Results and Discussion

We executed the ELECTRE TRI sorting algorithm using two different majority thresholds (λ) to test how strict the consensus needs to be. We calculated both Pessimistic and Optimistic assignments for each scenario.

4.2.1 Global Distribution Analysis

The histogram below provides a visual overview of how products are distributed across classes compared to the real Nutri-Score.

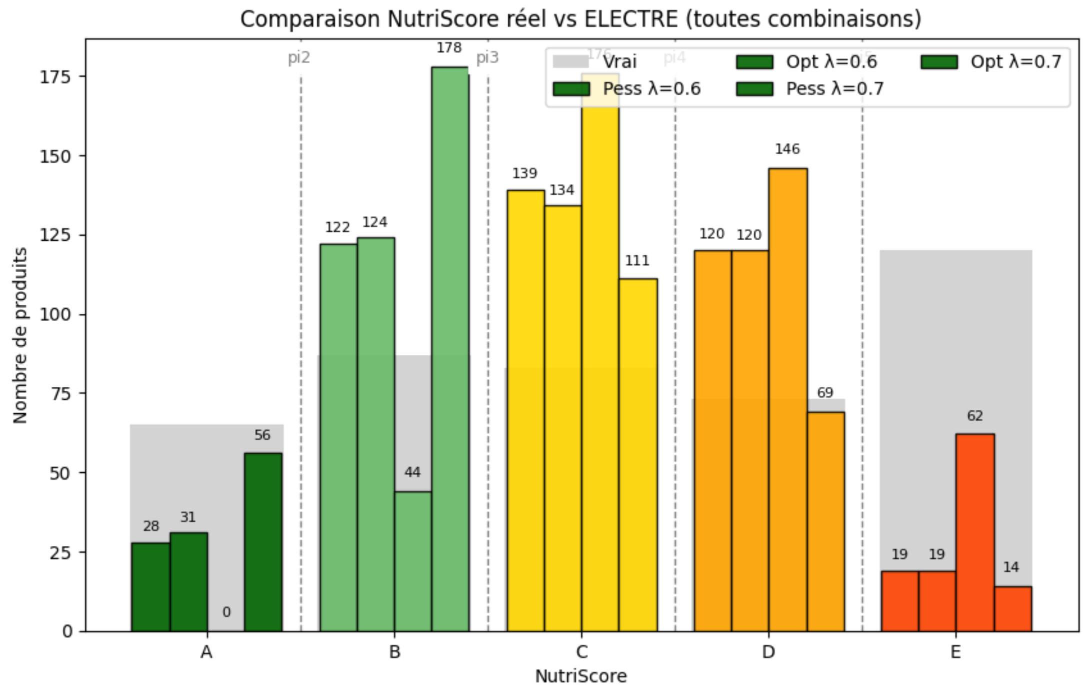

Figure 4: Distribution of products: Real Nutri-Score vs. ELECTRE Assignments.

4.2.2 Scenario A: Moderate Consensus ($\lambda = 0.6$)

In this first scenario, we require a simple majority (roughly 5 out of 8 criteria) to validate that a product is good enough for a category.

Looking at the confusion matrices for $\lambda = 0.6$:

- The Pessimistic assignment is relatively balanced but stricter than the real Nutri-Score.
- We observe a notable shift where products originally classified as 'A' are downgraded to 'B' or 'C'. This is largely due to the introduction of the Green-Score. A product might be nutritionally excellent but environmentally average, preventing it from passing the majority test for the top category.

4.2.3 Scenario B: Strong Consensus ($\lambda = 0.7$)

In this second scenario, we raised the bar. We require a strong majority (roughly 6 out of 8 criteria) to validate a classification.

The impact on the results is drastic:

- The Pessimistic model becomes extremely severe. As seen in the histograms, the number of products reaching category 'A' drops to near zero.
- Ideally, to be in class 'A', a product must now be excellent in almost every single dimension (Nutrition AND Environment). Very few industrial products in our database meet this high standard.

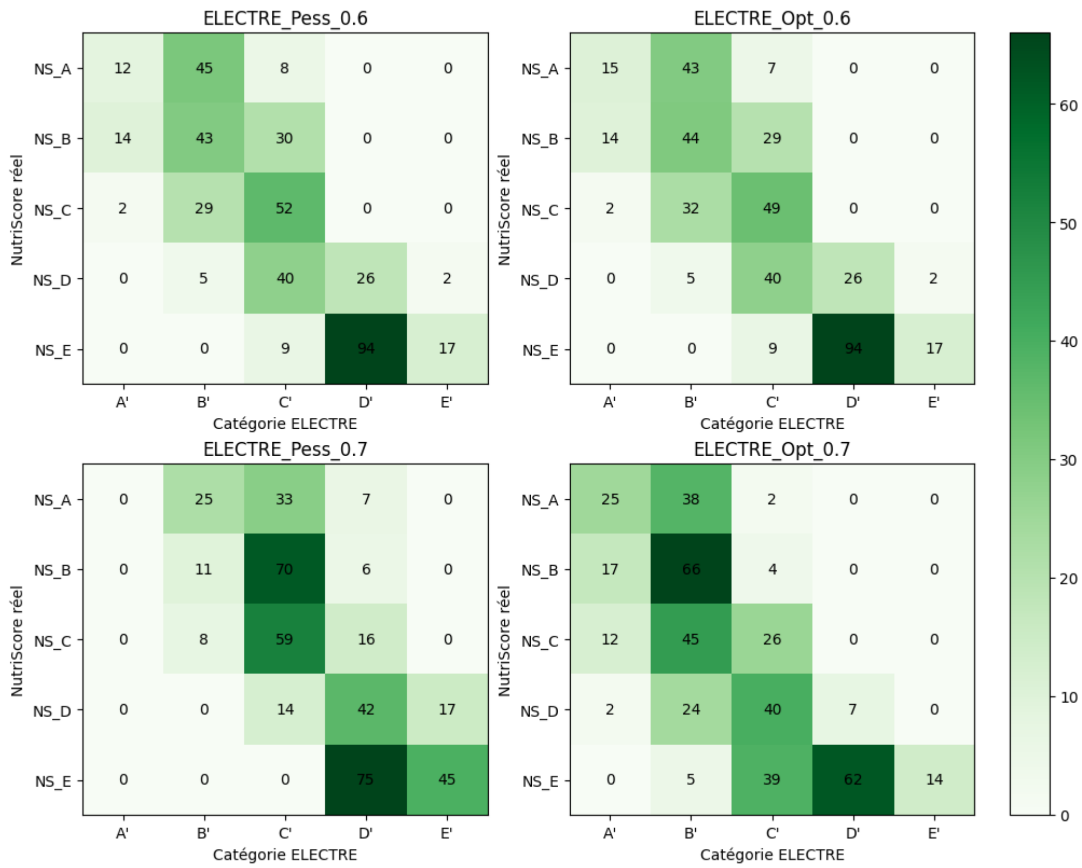

Figure 5: Confusion Matrices comparing Real Nutri-Score with ELECTRE predictions.

4.2.4 Comparison and Conclusion

Comparing the two thresholds reveals the sensitivity of the model. While $\lambda = 0.6$ offers a realistic alternative to the current Nutri-Score by gently integrating environmental concerns, $\lambda = 0.7$ acts as a very strict filter. This suggests that if policymakers want to create a "Super-Score" combining health and ecology, a moderate threshold is preferable to avoid discouraging consumers with too many low grades.

5 Weighted Sum Model (WSM)

5.1 Methodology and Formulation

As an alternative to the non-compensatory ELECTRE TRI method, we implemented a compensatory Weighted Sum Model (WSM). This approach allows a poor performance on one criterion (e.g., high sugar) to be offset by an excellent performance on another (e.g., very high Green-Score).

5.1.1 Normalization

Since the criteria have different units (grams, kJ, percentage), we applied a Min-Max normalization to scale all values between 0 and 1. To ensure consistency with the Nutri-Score logic (where a lower score implies a better grade), we normalized "Benefit" criteria (to maximize) and "Cost" criteria (to minimize) so that 0 represents the best performance and 1 represents the worst.

The normalization formulas used in our Python implementation are:

$$v_j(a) = \frac{g_j(a) - g_j^{min}}{g_j^{max} - g_j^{min}} \quad \text{(for Cost criteria, e.g., Sugar)} \tag{2}$$

$$v_j(a) = \frac{g_j^{max} - g_j(a)}{g_j^{max} - g_j^{min}} \quad \text{(for Benefit criteria, e.g., Fiber)} \tag{3}$$

5.1.2 Aggregation and Classification

The global score $S(a)$ is calculated as the weighted sum of the normalized values:

$$S(a) = \sum_{j=1}^{n} w_j \cdot v_j(a) \tag{4}$$

We utilized uniform weights ($w_j = 0.125$) for the 8 criteria. The final scores were stretched to fully cover the $[0, 1]$ interval. The classification into categories A to E was performed using linear thresholds with a step size of 0.2:

- **Class A:** Score in range $[0.0, 0.2]$
- **Class B:** Score in range $]0.2, 0.4]$
- **Class C:** Score in range $]0.4, 0.6]$
- **Class D:** Score in range $]0.6, 0.8]$
- **Class E:** Score in range $]0.8, 1.0]$

5.2 Results and Comparison

We applied this model to our dataset and generated the confusion matrix comparing the WSM predictions against the real Nutri-Score.

Figure 6: Confusion Matrix of the Weighted Sum Model (WSM).

Interpretation of Results: The WSM demonstrates a different behavior compared to ELECTRE TRI:

1. **Central Tendency:** The model tends to cluster predictions around the middle classes. We observe that 39 products originally labeled 'A' were downgraded to 'B', and 55 products labeled 'C' were upgraded to 'B'.
2. **Compensatory Effect:** Unlike ELECTRE, where a single bad criterion (veto effect) can block a product from a high class, WSM allows averaging. This explains why the diagonal accuracy for extreme classes is lower. The inclusion of the Green-Score pulls "purely nutritional" A-products down if their environmental impact is only average.
3. **Comparison with ELECTRE:** The WSM is computationally simpler but less discriminatory for top-tier products. It fails to maintain the strict requirements needed for the 'A' label, often merging distinct profiles into the 'B' or 'C' categories.

6 Machine Learning Model: Random Forest

6.1 Approach and Configuration

To address the challenge of effectively combining the Nutri-Score and the Eco-Score using Artificial Intelligence, we implemented a Random Forest Classifier. Unlike the ELECTRE or Weighted Sum models, which rely on explicit decision rules defined by humans, this model learns classification patterns directly from the data.

We configured the model with the following parameters:

- **Number of trees:** 200 estimators.
- **Criteria:** The same 8 attributes used previously (Nutrients + Green-Score).
- **Target:** The Nutri-Score class (A to E).

A key feature of our approach is the training strategy. We employed a stratified split with a very small training set (only 50 products) to rigorously test the model's ability to generalize to the rest of the database (approximately 370 test products).

6.2 Analysis of Results

6.2.1 Feature Analysis (Dendrogram)

Before evaluating performance, we generated a hierarchical dendrogram to visualize how the model perceives the relationships between different criteria.

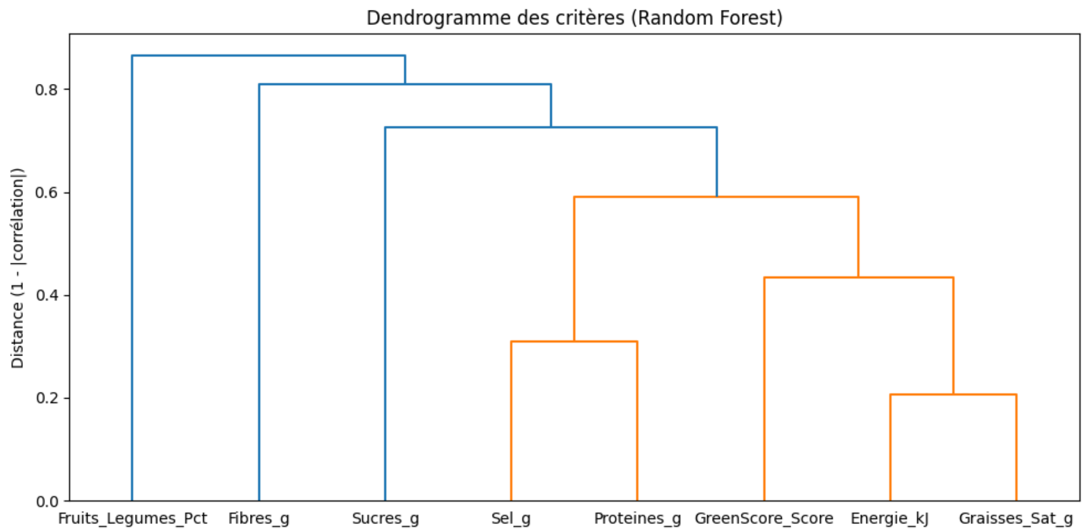

Figure 7: Dendrogram of criteria showing correlations.

The analysis of Figure 7 reveals logical clusters:

1. On the left, we observe a strong link between Fruits/Vegetables and Fibers, which makes biological sense.
2. On the right, Energy and Saturated Fats form another distinct cluster.
3. The Green-Score is positioned somewhat independently, confirming that it provides unique information to the model that is not redundant with simple nutritional values.

6.2.2 Classification Performance

The confusion matrix below illustrates the model's predictions on the test set (products the model had never seen before).

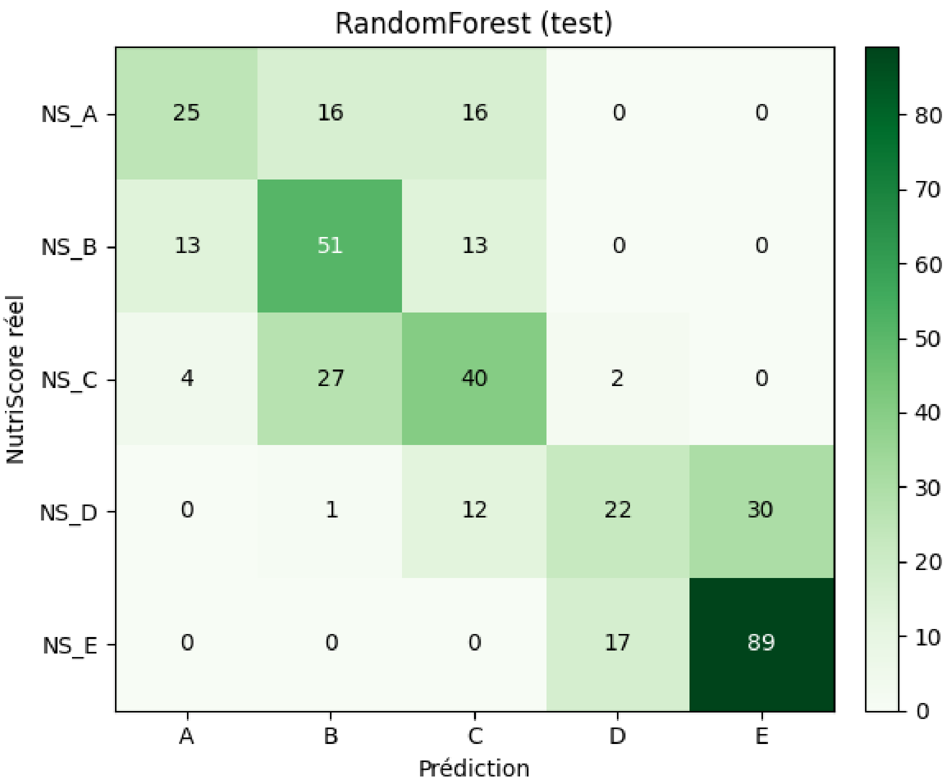

Figure 8: Confusion Matrix of the Random Forest on the test set.

As shown in Figure 8, the Random Forest achieves excellent results, outperforming the previous models:

- The diagonal is very prominent, indicating a high success rate.
- Class E is detected particularly well (89 correct predictions, with very little confusion).
- Errors remain close to the diagonal (for example, a few "C" products classified as "B" or "D"), showing that the model does not make aberrant errors.

6.3 Conclusion on Machine Learning

This model demonstrates that it is possible to replicate the classification logic with high precision, even with limited training data. However, although it is more accurate than the ELECTRE method, the Random Forest acts as a "black box": it is harder to explain to a consumer exactly why a product lost a grade, unlike the explicit threshold approach.

7 Final Model Recommendation

To conclude this project, we evaluated which of the developed models (ELECTRE TRI, Weighted Sum, or Random Forest) is the most suitable for implementing a combined Nutri-Eco-Score.

7.1 Comparative Assessment

We judged the models based on three criteria:

- Accuracy: The Random Forest was the most accurate in replicating existing patterns, achieving high predictive success on the test set.
- Transparency: ELECTRE TRI offers the best explainability. We can trace exactly why a product failed to reach a higher class (e.g., due to a specific threshold in salt or Green-Score).
- Safety: The Weighted Sum Model is risky for health labeling because it allows a good environmental score to compensate for poor nutritional quality, which is undesirable.

7.2 Our Choice: ELECTRE TRI

Among all models, we are most comfortable recommending the ELECTRE TRI model (specifically with lambda = 0.6).

Although the Random Forest had higher statistical accuracy, it operates as a "black box". In the context of public health and consumer information, trust is built on explainability. A consumer needs to understand that a product is rated "B" instead of "A" specifically because of its environmental impact, not because of an opaque algorithmic calculation.

Furthermore, ELECTRE's non-compensatory nature ensures that a product cannot be labeled as excellent if it has a disqualifying defect in either nutrition or ecology. The threshold of 0.6 provides a realistic balance, integrating the Green-Score without being excessively punitive.

8 Cross-Validation: Comparison with External Data

To evaluate the robustness of our decision model, we applied our ELECTRE TRI algorithm to a dataset provided by another student group. The objective was to determine if the classification patterns we observed internally—specifically the strictness introduced by the Green-Score—would persist on a completely new set of products.

8.1 Visual Comparison of Results

The figure below juxtaposes the results obtained on our own dataset (Left) against the results obtained on the other group's dataset (Right), using our Pessimistic sorting method.

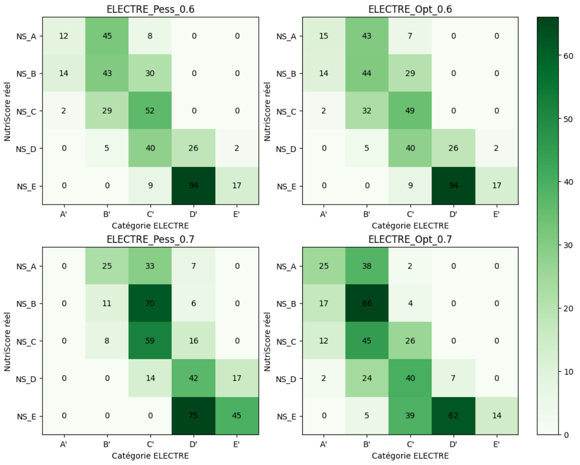

Figure 9: Results on **Our Dataset**

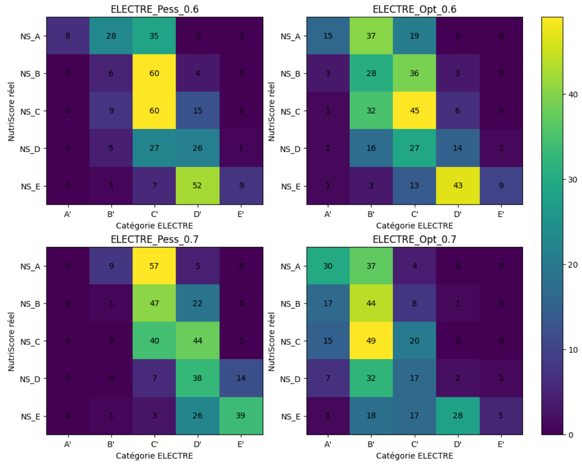

Figure 10: Results on **External Dataset**

8.2 Analysis of Trends ($\lambda = 0.6$ vs $\lambda = 0.7$)

By analyzing the behavior of the model across both datasets, we can draw definitive conclusions about its structural impact.

8.2.1 Scenario $\lambda = 0.6$: The "Green Filter" Effect

When using a moderate consensus threshold of 0.6, we observe a striking similarity between the two groups. In both cases, the model acts as a filter for environmental quality.

- **Internal Data:** We initially noticed that many products rated 'A' by the standard Nutri-Score were downgraded to 'B' or 'C'.
- **External Data:** This pattern is perfectly replicated on the other group's data. The confusion matrix shows a dispersion from the diagonal for the best-rated products.

This confirms that the "downgrading" effect is not an anomaly of our specific database. It is a systematic feature of our model: a product cannot remain in category 'A' if its Green-Score is merely average, regardless of the dataset origin.

8.2.2 Scenario $\lambda = 0.7$: The "Elitist" Effect

When raising the threshold to 0.7, the model becomes extremely severe for both datasets.

- **Consistency in Strictness:** In our results, the 'A' category became almost empty. Similarly, on the external dataset, the model refused to grant the 'A' label to the vast majority of products.
- **Interpretation:** This validates that a consensus of 70% across 8 criteria is nearly impossible to reach for standard industrial food products. The fact that this occurs in both groups proves that our limiting profiles are not overfitted; they simply reflect a high standard of excellence that few current products meet.

8.3 Conclusion on Robustness

The comparison proves that our ELECTRE TRI model is robust. It applies the same logic consistently: it sanctions products with poor environmental performance and requires a near-perfect profile for the top categories. The similarity in the confusion matrices demonstrates that our model is a reliable, albeit strict, tool for decision-making.

9 Conclusion

In conclusion, this project successfully demonstrated that integrating environmental criteria into the Nutri-Score is methodologically feasible but requires careful model selection. While our experiments showed that Machine Learning offers the highest predictive accuracy and the Weighted Sum Model provides mathematical flexibility, we ultimately recommend ELECTRE TRI as the superior approach for public policy. Its non-compensatory nature ensures that a good environmental score cannot mask poor nutritional quality, while its threshold-based logic maintains the transparency essential for consumer trust. By adopting this multi-criteria framework, we can effectively evolve the Nutri-Score into a comprehensive tool that guides consumers toward choices that are both healthy for the body and sustainable for the planet.